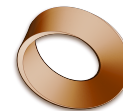


Name: _____

Date: _____

Mr. Rawson • Y8 MYP Physics



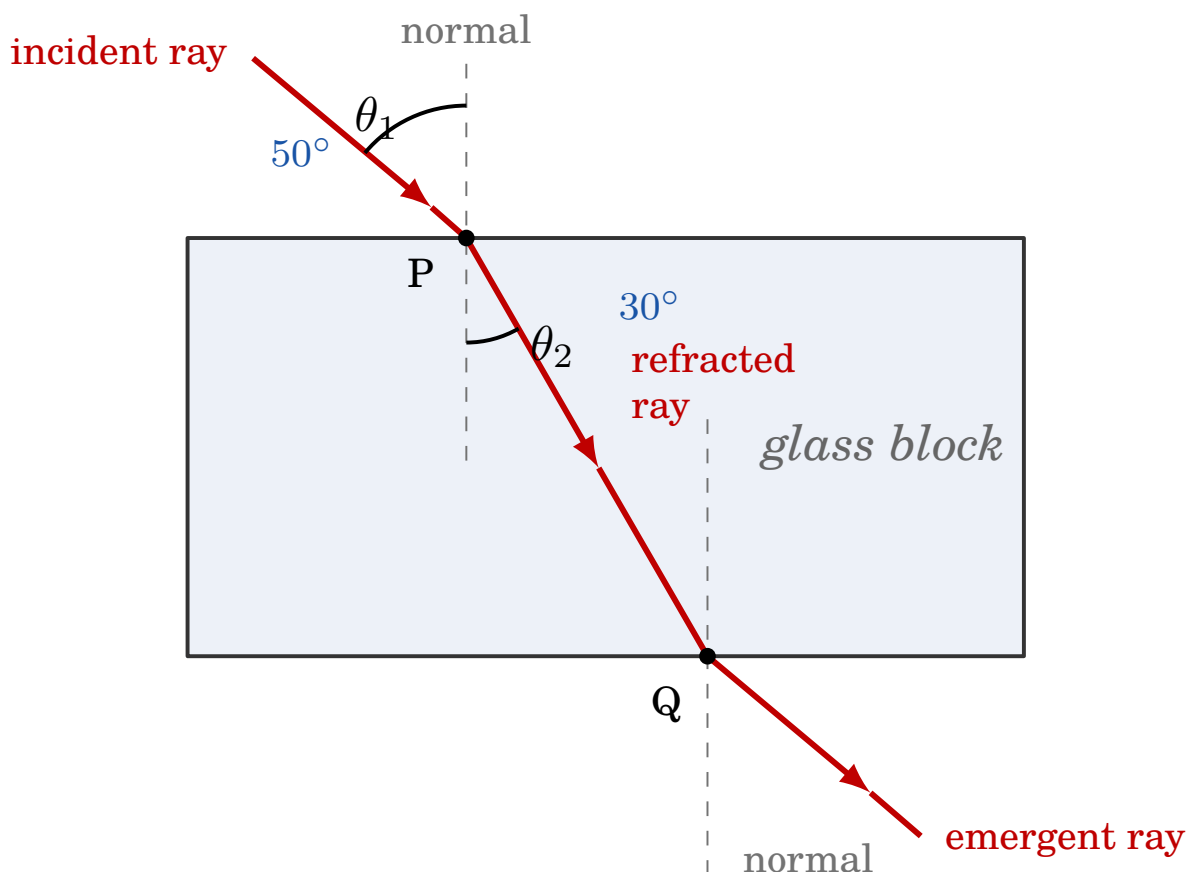
1.5 – Measuring Refraction

Unit 1: Light & EM Spectrum

Learning intentions

Last lesson we marched the boundary ourselves and found out **why** light bends. Today we find out **how much** it bends — and we learn the way of measuring it that you will need for your investigation.

Light is travelling through air and meets a **glass block**. Glass slows light down, so the ray bends **towards the normal** as it goes in. At the far side it speeds up again, and bends **away** from the normal on the way out.

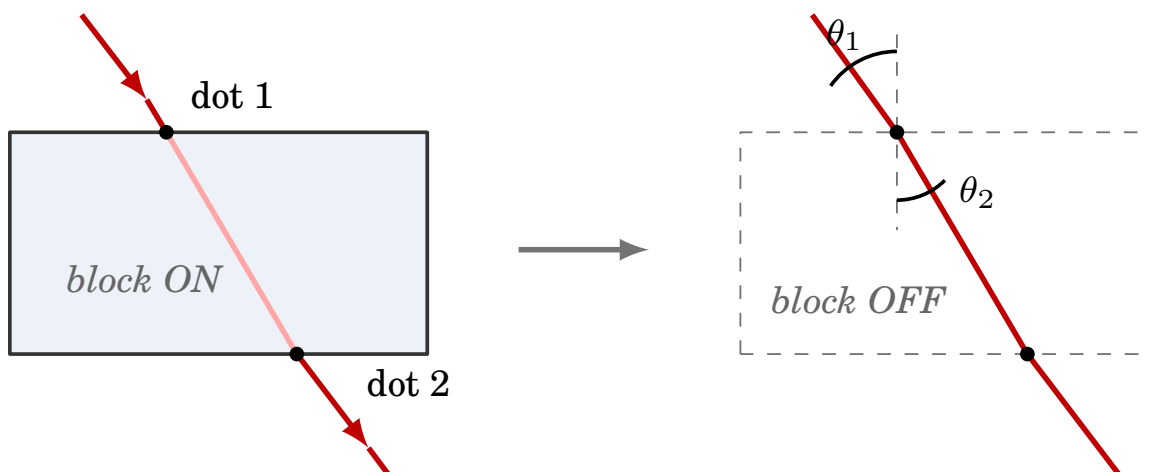


Two things to notice on that diagram before you touch any apparatus.

- The ray bends at **P** and at **Q** — at the two **boundaries**, and nowhere else. In between it is straight.
- The **emergent ray** is **parallel** to the incident ray. The block shifts the light sideways, but it does not turn it.

1. The method

You have a **ray box**, a **rectangular glass block**, a **protractor**, a **ruler**, and a **sharp pencil**. Work in **pencil** throughout — you will want to rub things out, and that is normal.



1. Mark BOTH dots.

**2. Lift the block. Join the dots.
Draw the normal. Measure.**

1. Put the glass block flat on a plain sheet of paper and **draw round it** with a sharp pencil. Do not move it after this.
2. Shine a single ray at the block so it hits the long face **at an angle**, not straight on.
3. Mark the ray with **two dots**: one where it **goes in**, one where it **comes out**.
4. **Now** lift the block off. Join your two dots with a ruler — that line is the path the light took *inside* the glass.
5. At the point where the ray went in, draw the **normal**: a line at exactly 90° to the surface of the block.
6. Measure θ_1 and θ_2 **from the normal**, with the middle of your protractor on the dot.

Before you lift the block

Check you have **both** dots. Once the block has moved you cannot put it back in the same place, and the whole trace has to be started again.

One thing to watch, and it caught people out in the reflection practical. Every angle here is measured from the **normal**, never from the **surface** of the glass. If your two angles add up to about 90° , you have measured from the surface — turn your protractor and measure again.

2. Your results

Repeat the method for each angle of incidence in the table. Measure θ_2 each time, to the nearest degree.

Angle of incidence θ_1 (in air)	Angle of refraction θ_2 (in glass)	Did the ray bend towards or away?
20°	$\approx 13^\circ$	towards
30°	$\approx 19^\circ$	towards
40°	$\approx 25^\circ$	towards
50°	$\approx 31^\circ$	towards
60°	$\approx 35^\circ$	towards

Values for $n = 1.5$ glass. Anything within a couple of degrees is a good measurement; what matters is that $\theta_2 < \theta_1$ every time and that the gap widens as θ_1 grows.

3. What your results show

(a) Compare each pair of angles. Is θ_2 always bigger than θ_1 , always smaller, or does it vary?

θ_2 is always **smaller** than θ_1 . Going from air into glass the ray always bends towards the normal.

(b) Use the word **speed** to explain why.

Light travels **slower** in glass than in air. The side of the ray that reaches the glass first slows down first, while the other side is still going quickly — so the ray swings round towards the normal. (This is the marching band crossing onto the long grass.)

(c) What would happen to θ_2 if you used a block of **diamond** instead of glass? ($n_{\text{glass}} = 1.50$, $n_{\text{diamond}} = 2.42$)

θ_2 would be even **smaller**. Diamond has a bigger n , so light is slower in it, so the ray bends more towards the normal.

(d) A student shines the ray so that it hits the block **straight on**, along the normal. Predict what the ray does.

It carries straight on without bending. $\theta_1 = 0^\circ$, so no part of the ray reaches the glass before any other part — nothing swings round. It still slows down, but it does not change direction.

Extension

For each row of your table, work out $\theta_1 \div \theta_2$.

Then work out $\sin \theta_1 \div \sin \theta_2$.

One of those two calculations gives roughly the **same answer every time**. Which one — and what is that number close to?

$\sin \theta_1 \div \sin \theta_2$ is the constant one, and it comes out at about **1.5** — the index of refraction of glass. Dividing the angles themselves does not give a constant. (This is Snell's law, and you have just measured it.)